General Chemistry I

Chemistry: The central science

Phong D. Tran / Anh D. Nguyen

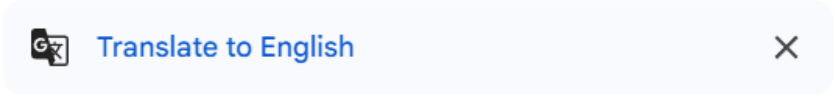
University of Science and Technology of Hanoi

Email: tran-dinh.phong@usth.edu.vn;

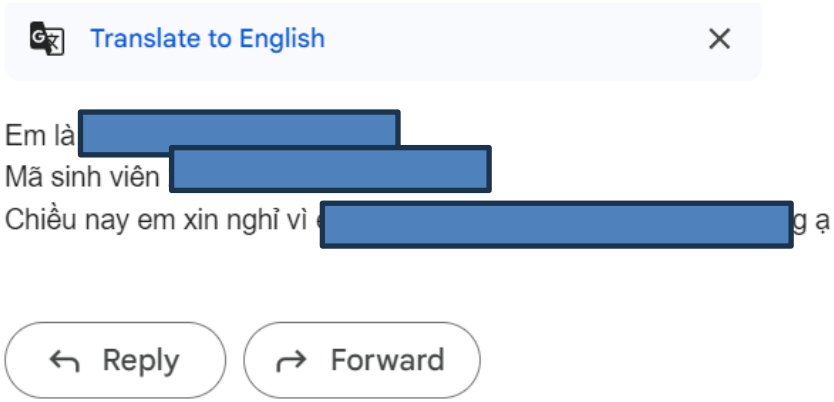
nguyen-duc.anh@usth.edu.vn

Office: 305 – 716, USTH A21 Building

Lab: 703-707, USTH A21 Building



Dear .....



Best regards / Sincerely yours,

Education



2003
Bachelor of Chemistry
VNU Hanoi

2008
PhD, Inorganic Chemistry
Universite Paris Sud

2016
HDR, Chemistry
Universite Grenoble Alpes

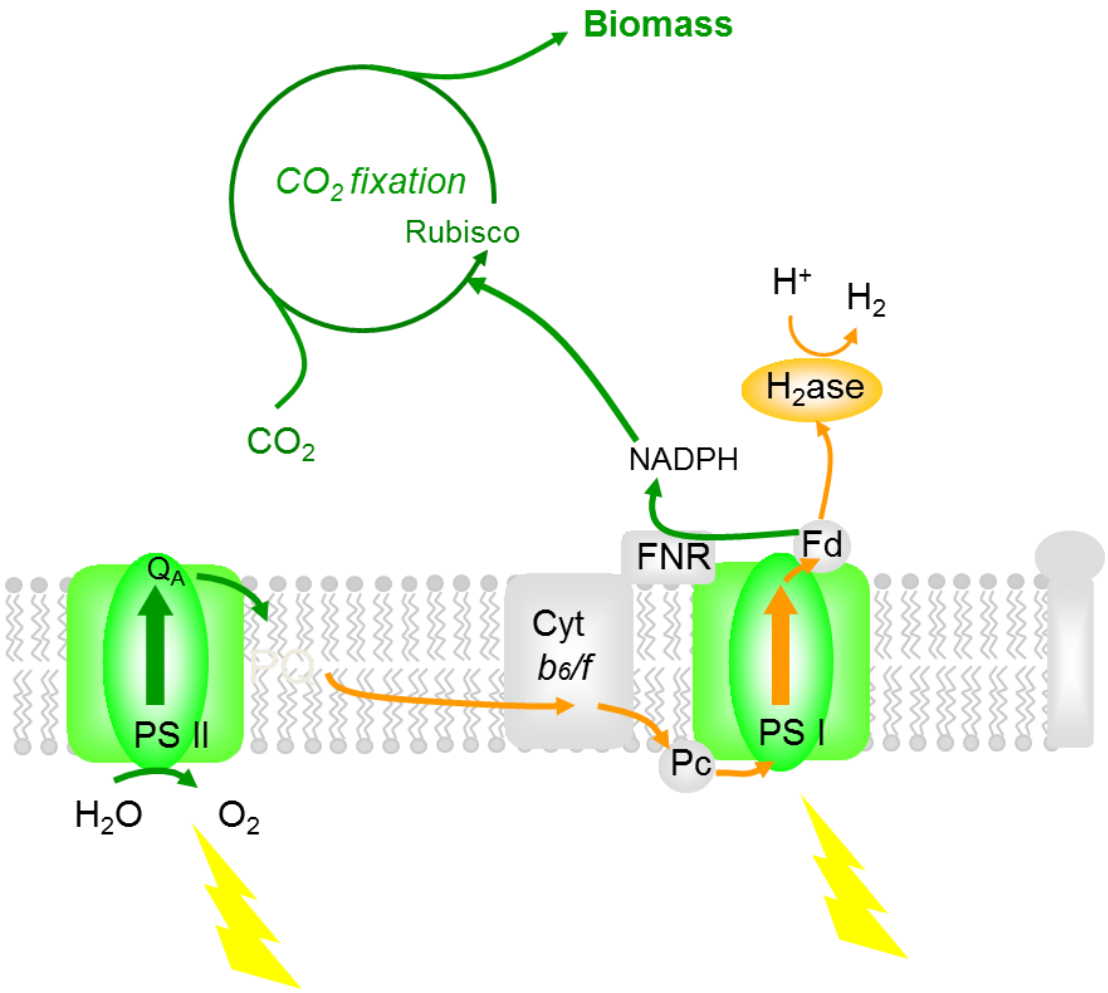


2009
Engineer of Polymer Technology
School of Chemistry and Life Sciences ,
HUST

2016
PhD, Chemistry of Polymers
Université du Maine

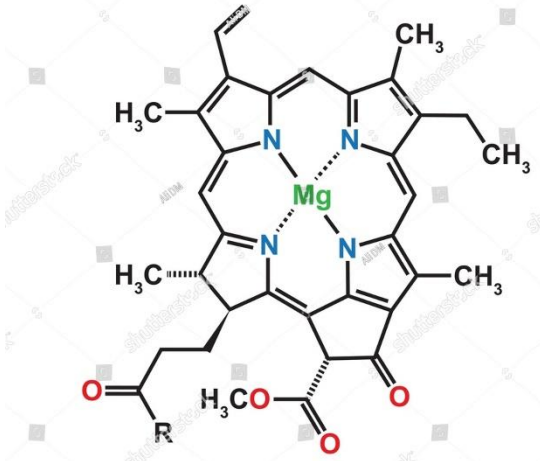
Natural photosynthesis:

A fascinating machinery for CO_2 fixation and solar-to-chemical energy conversion

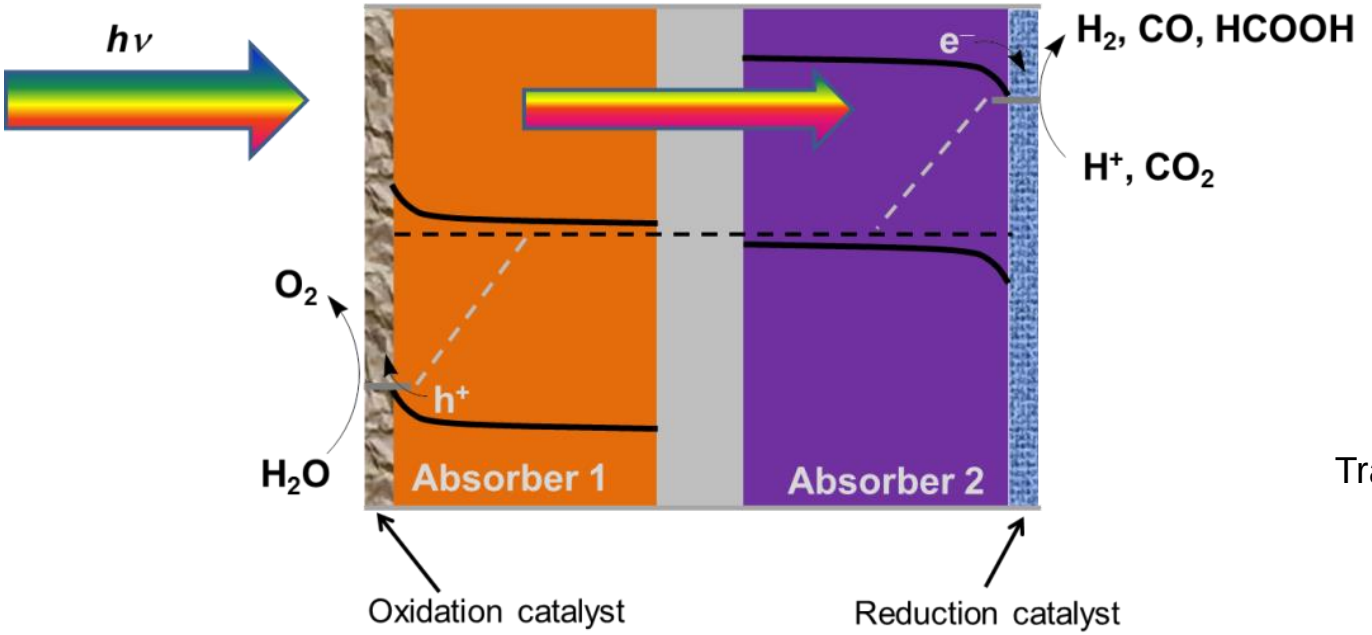


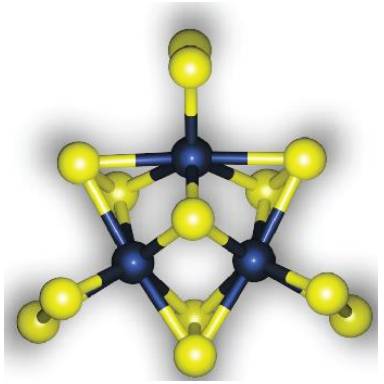


Chlorophyll a $C_{55}H_{72}O_5N_4Mg$



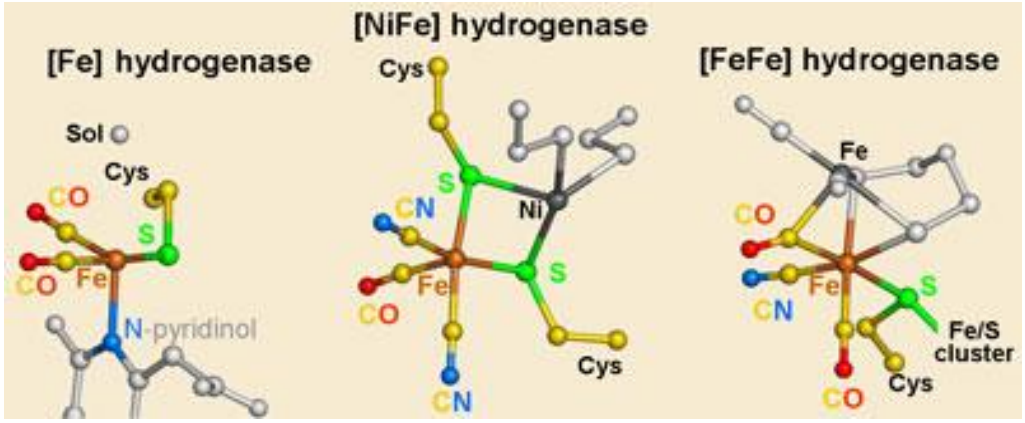
Artificial Leaf

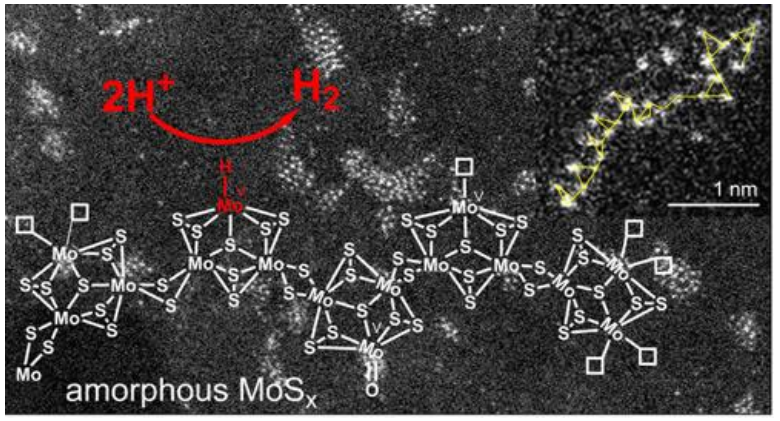




Tran et al., *Nature Materials*, 2016, **15**, 640–646

Catalyst for H_2 generation





Our first Artificial leaf



Nguyen et al., *J. Am. Chem. Soc.* **2022**, 144, 9651-9660

Laboratory of Chemistry for Energy Conversion & Storage



@ **Date of creation**: July 2015

@ **Current staffs**

* 07 Permanent lecturer-researchers
* 01 Lab secretary
* 02 Postdoc
* 06 PhD students
* 02 Master students

@ **Publication: 6-8 SCIE (Q1)/ year**
(Nature Materials, JACS, ACS Catalysis, Chemical Communication, *etc*)

@ **Alumni:** >40 (USA, FR, POL, AUS, JP, KR)

Funding support

VAST

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RÉPUBLIQUE FRANÇAISE
AMBASSADE DE FRANCE
AU VIETNAM

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USTH
VIETNAM FRANCE UNIVERSITY

CONSORTIUM USTH
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Office of Naval Research Global
Science & Technology

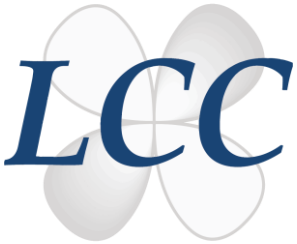
HỌC VIỆN KHOA HỌC VÀ CÔNG NGHỆ
GRADUATE UNIVERSITY OF SCIENCE AND TECHNOLOGY

VINIF

CECS: Current Collaboration



Dr. Vincent Artero, UMR 5249



Prof. Catherine Amiens
UPR 8241



Prof. Gilles Lemercier, UMR 7312



Prof. Hyuksu Han
(Konkuk Univ.)



Prof. Iratu Honma
(Univ. of Tohoku)



Dr. Chiam Sing Yang
(IMRE @ ASTAR)



Prof. Nguyen Quang Liem
(IMS, VAST)

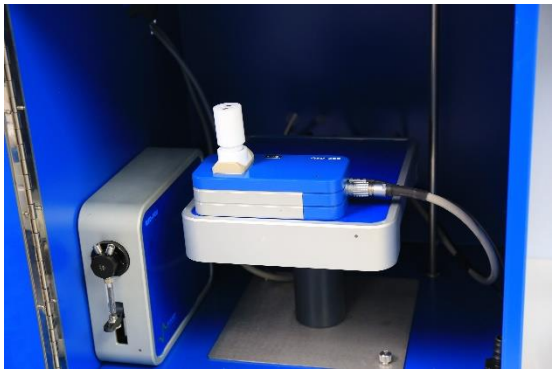


Prof. Luc Huy Hoang
(HNUE)



Dr. Luu Anh Tuyen
(VINATOM)

Equipment available in CECS





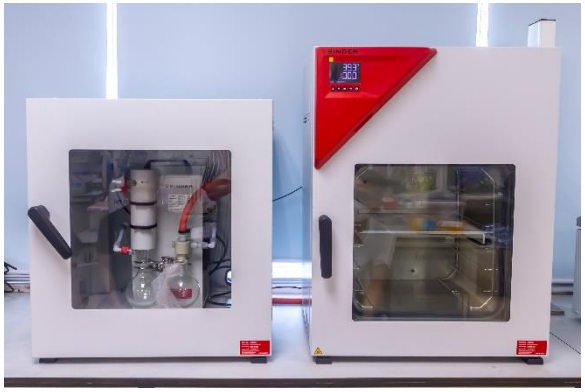














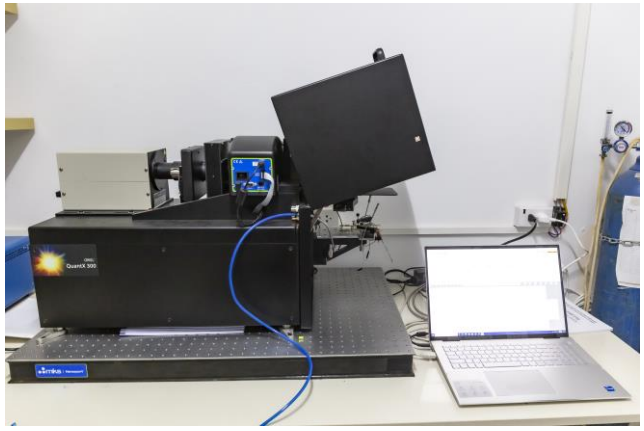
…and more

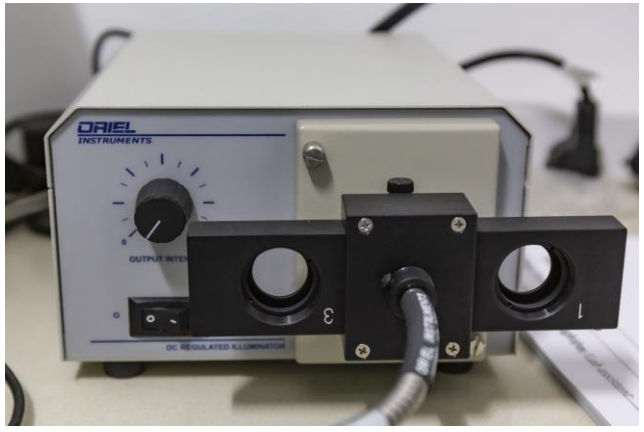


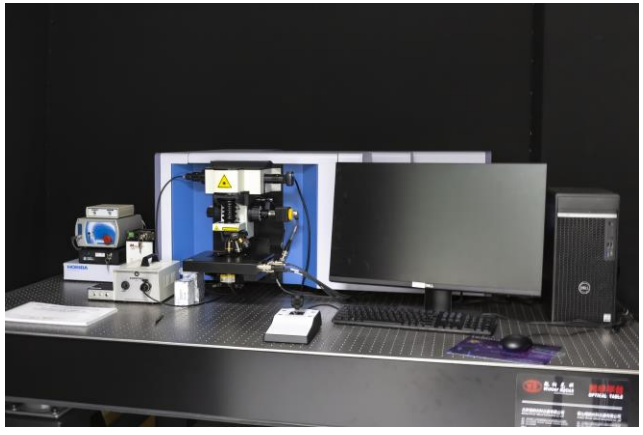


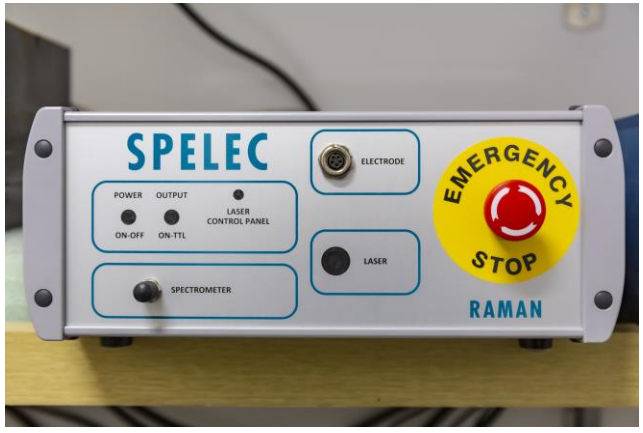

CECS
Agilent
Cary
UV-Vis Compact
31/12/2021






ORIEL
INSTRUMENTS




SPELEC
ELECTRODE
POWER
OUTPUT
ON-OFF
ON-TTL
LASER
CONTROL PANEL
SPECTROMETER
EMERGENCY
STOP
RAMAN


ELECTRODE
DEUTERIUM HALOGEN SHUTTER
ON-OFF
ON-OFF
ON-OFF-TTL
SPELEC
LIGHTSOURCE
LAMP
POWER
SPECTROMETER

USTH Alumni-2023







Please join us to build up a competitive research group



Rules in General Chemistry Class

* No Personal Stuffs (Music, Facebook, Phone, Google, *etc*)

** Actively involve in discussion/ exchange

*** Attendance checking: Random 50 students in class

(if you absent 5 sections → you then redo the class ... next year)

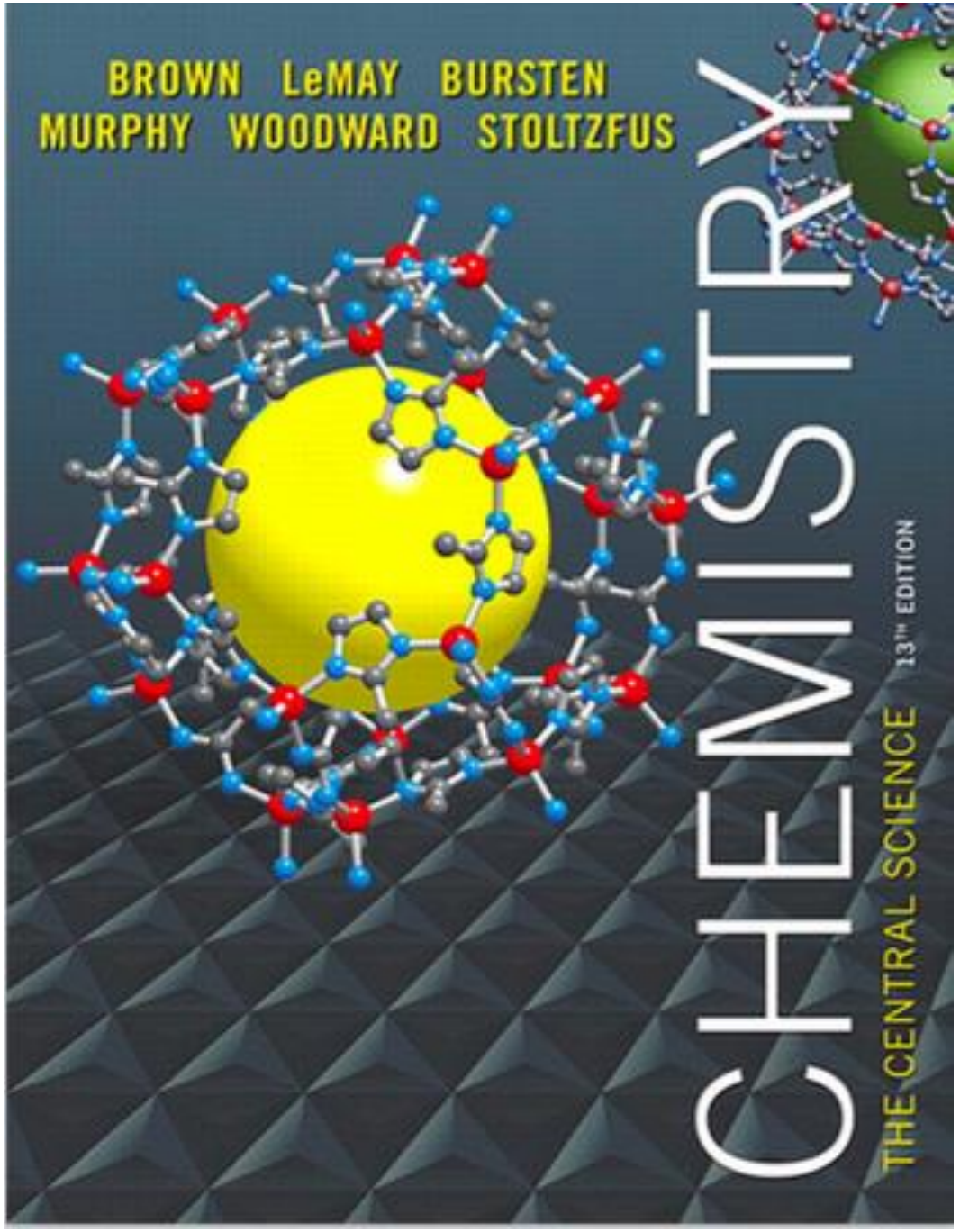
* Teaching:

Important concepts will be provided/ explained

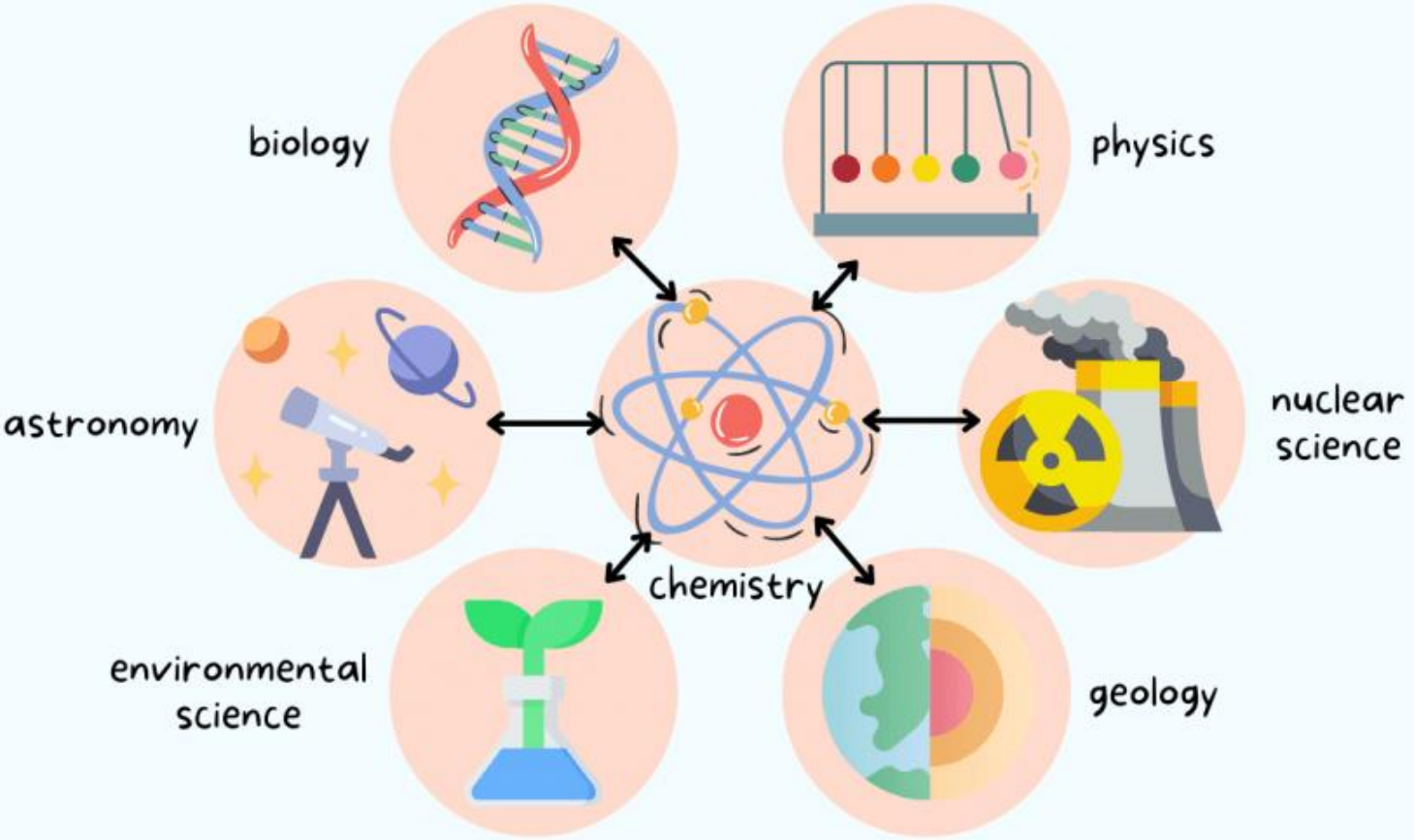
Lecture instruction → Students work with text-book

Understand and **Apply your knowledge** RATHER than ***Remember*** the information provided

The Text Book



Why Is Chemistry Called the Central Science?



Chemistry is the central science because its concepts are at the core of other sciences and applied sciences.

sciencenotes.org

General Chemistry I 28h lecture (14 sections)

Part I: Introduction

1. General Introduction (*Role of Chemistry*): Section 1
2. Matter and Measurement: Sections 1, 2

Part II: Structure of Atom and Molecule

1. Atoms (atomic structure, electronic structure of atoms): Sections 3-5
2. Periodic properties of elements: Sections 6,7
3. **Midterm exam: Section 8**
4. Bonding theories: Sections 9,10
5. Molecular geometry: Sections 11,12
6. Chemical reactions: Section 13
7. Final Revision: Section 14

+ 06 Tutorial sections

Total score`s constitution: **Attendance + Contribution (10%)**
Midterm (40%);
Final Exam (50%)

Question?

Comment?

Concern?



What is Chemistry/ Chemical Science?

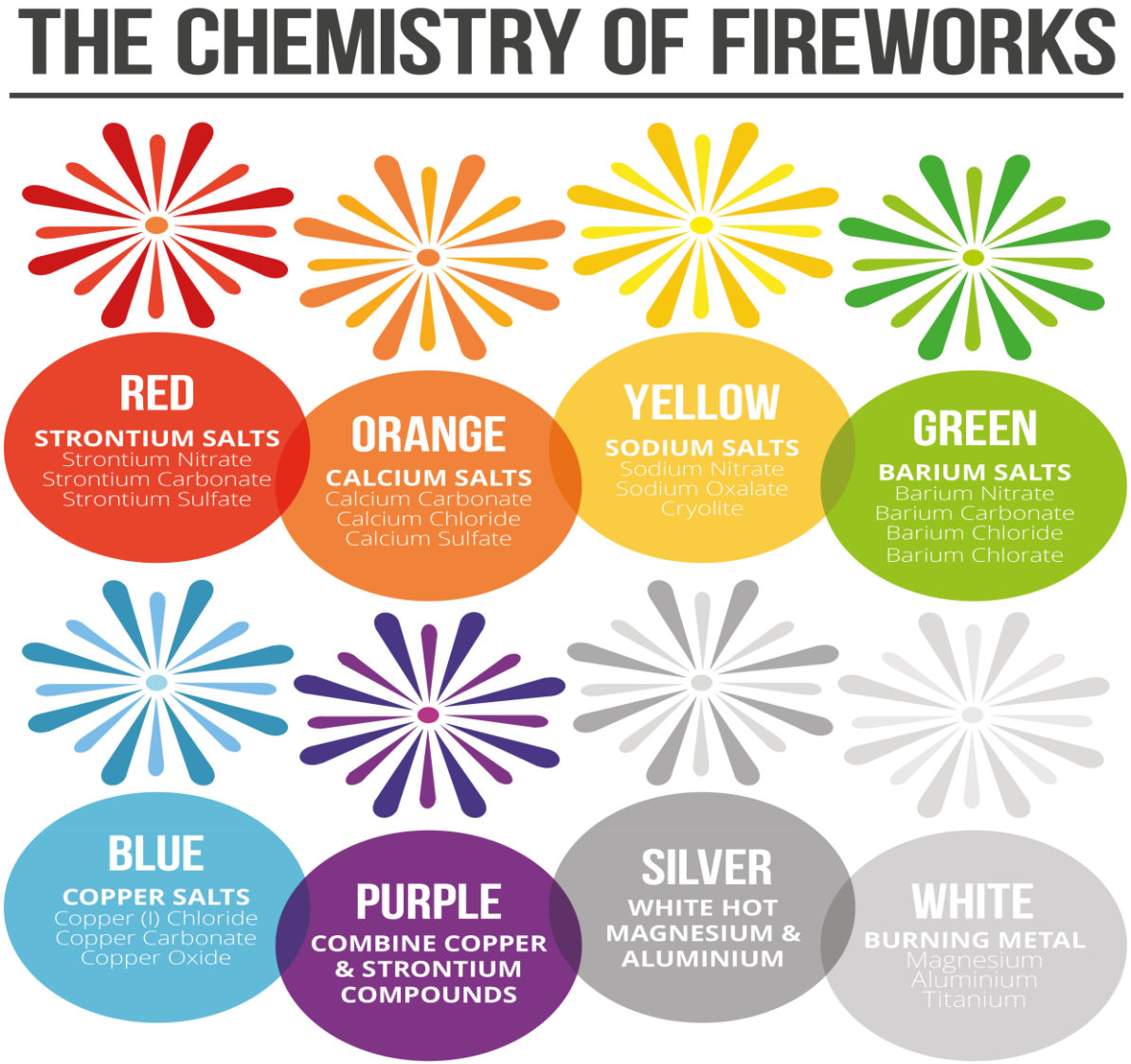
Why is Chemistry/ Chemical Science important?

How is Chemistry/ Chemical Science Easy?

What can we expect from this General Chemistry lecture?

What is Chemistry/ Chemical Science?

Chemistry is about color, smell, … About daily phenomenon?



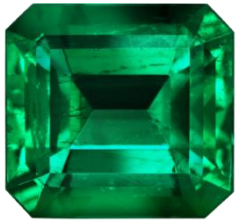
Chemistry of gemstone colors



RUBY
Al_2O_3



SAPPHIRE
Al_2O_3



EMERALD
$Be_3Al_2(SiO_3)_6$



ALEXANDRITE
Al_2BeO_4



AMETHYST
SiO_2



DIAMOND
C



GARNET
$Mg_3Al_2(SiO_4)_3$



TOPAZ
$Al_2SiO_4(F,OH)_2$



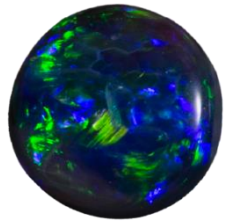
TOURMALINE
$Na_3Li_3Al_6(BO_3)_3(SiO_3)_6F_4$



ZIRCON
$ZrSiO_4$



TURQUOISE
$Al_6(PO_4)_4(OH)_8.4H_2O$



OPAL
$SiO_2.nH_2O$



PERIDOT
Mg_2SiO_4



PEARL
$CaCO_3$



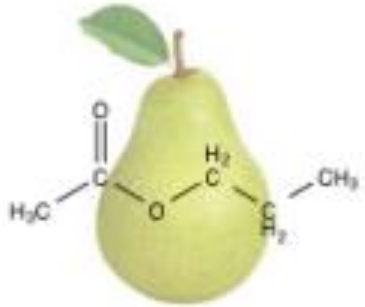
AQUAMARINE
$Be_3Al_2(SiO_3)_6$

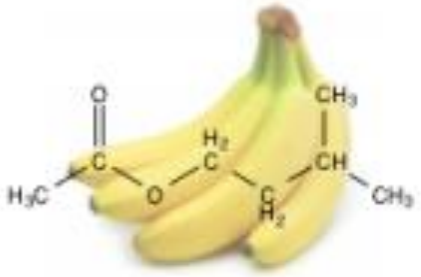


SPINEL
$Mg_3Al_2O_4$

The smells of fruits



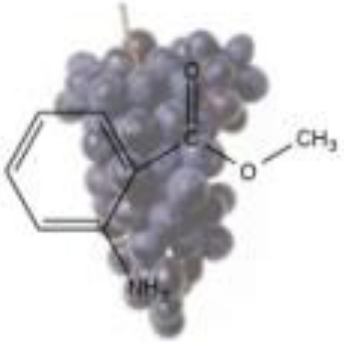


















Welcome to France

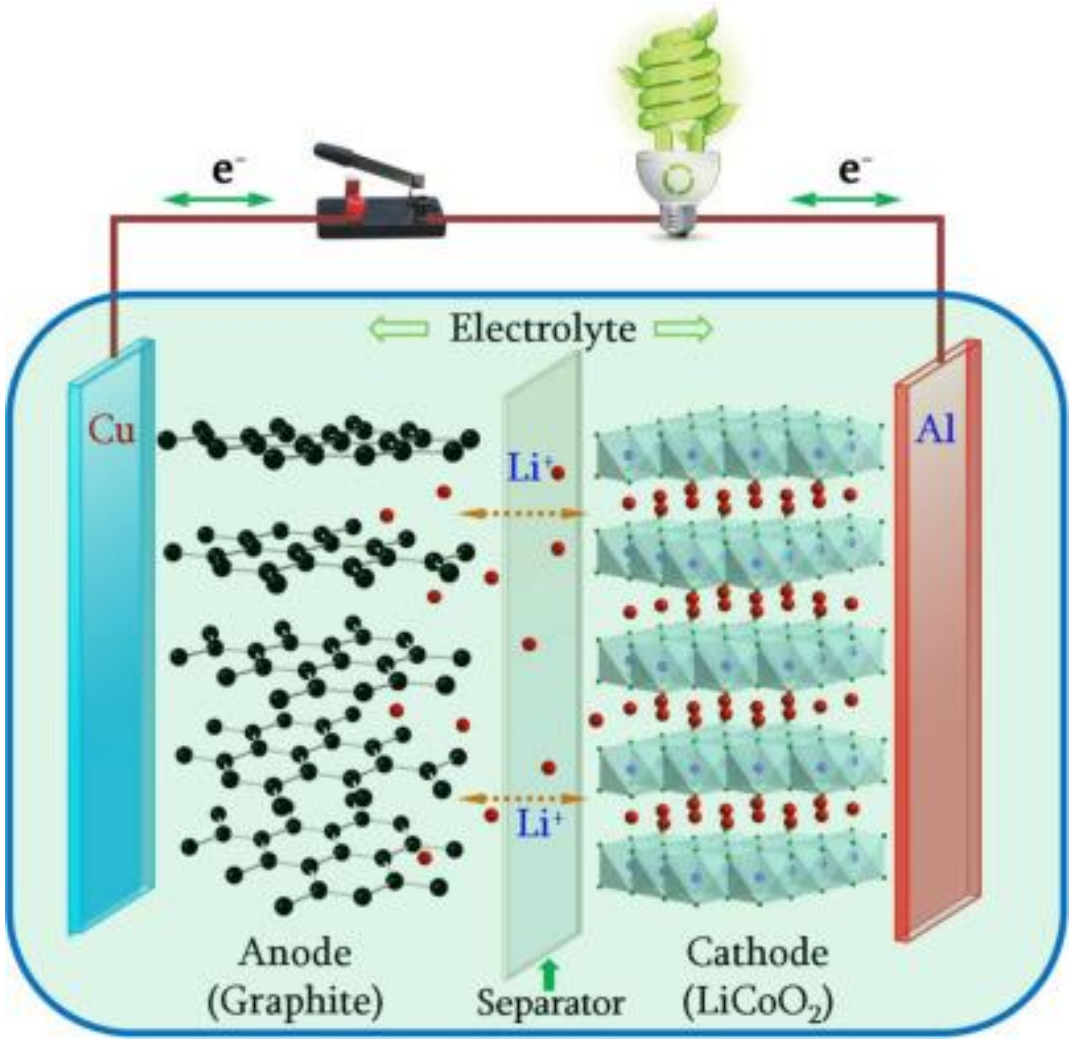




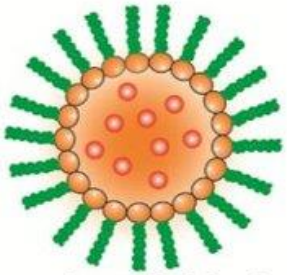
Li battery technology



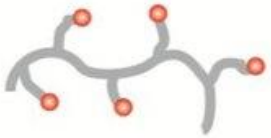




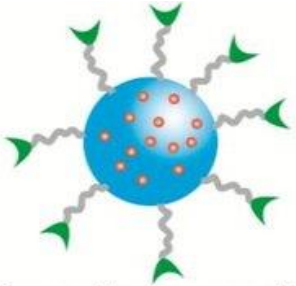
Drug Delivery technology



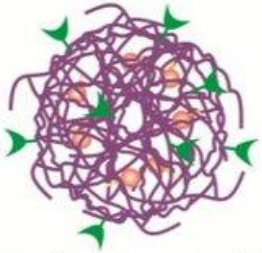
Polymeric or lipid micelles



Polymer-drug conjugates



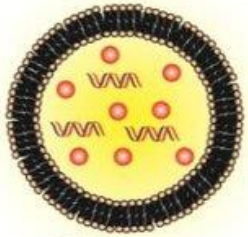
Polymeric nanoparticles



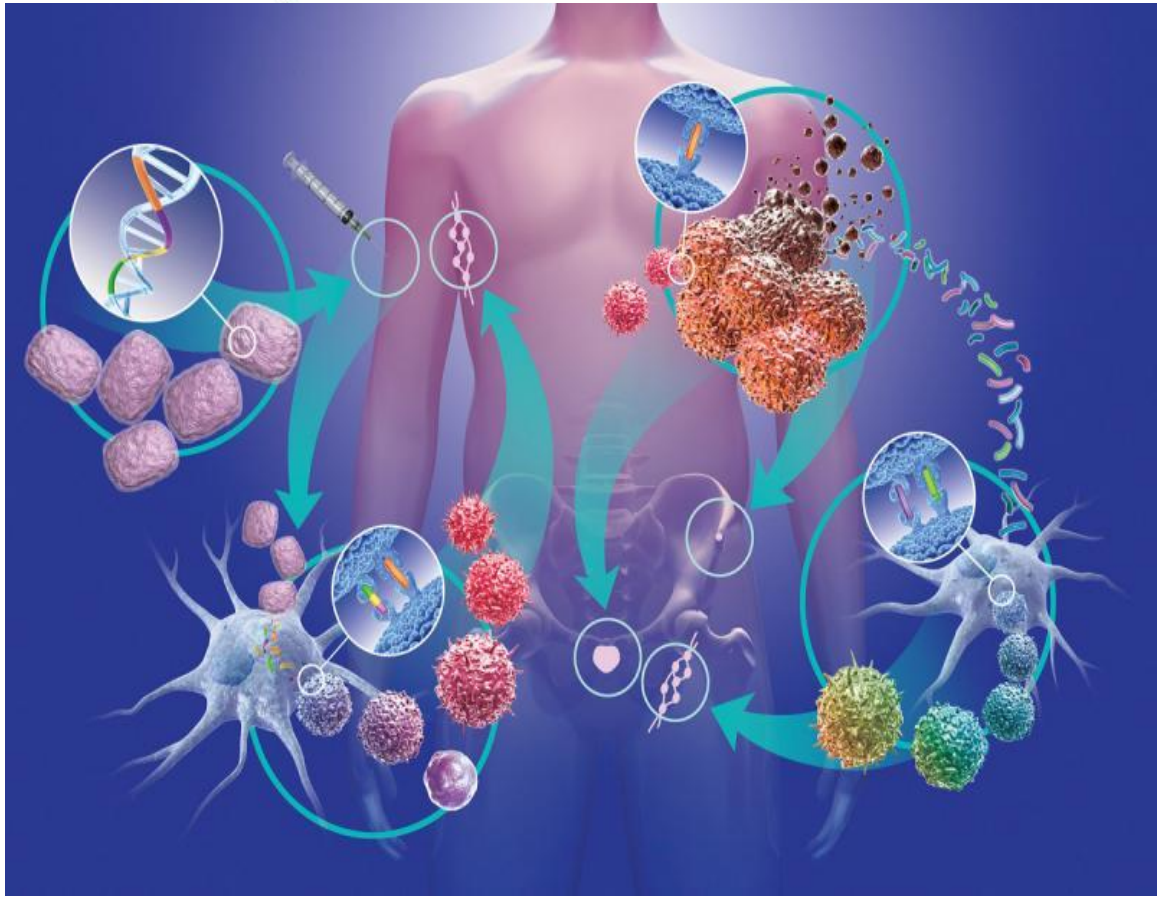
Protein nanoparticles



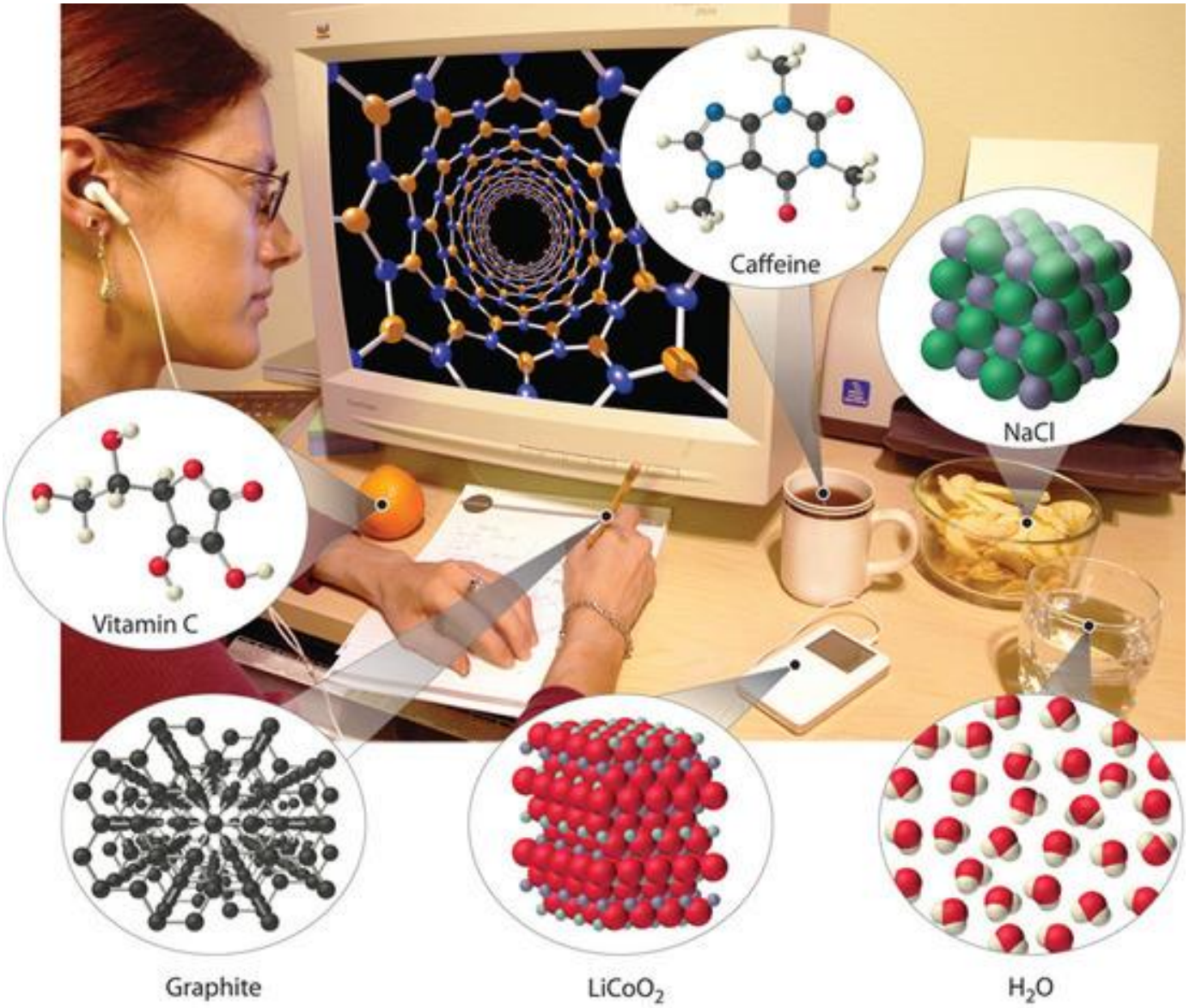
Dendrimers



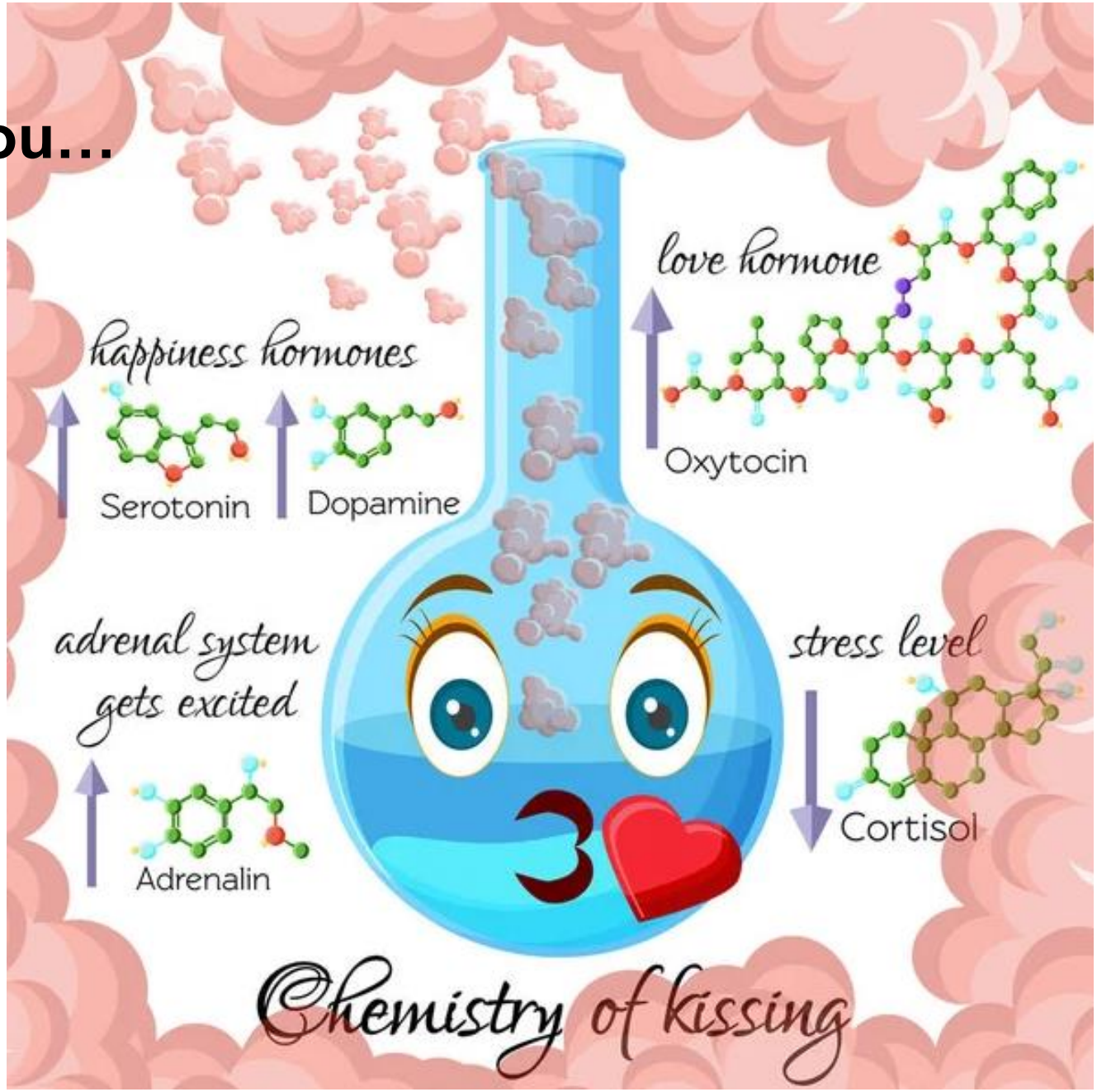
Liposomes



Chemicals are surround you…



And inside you…



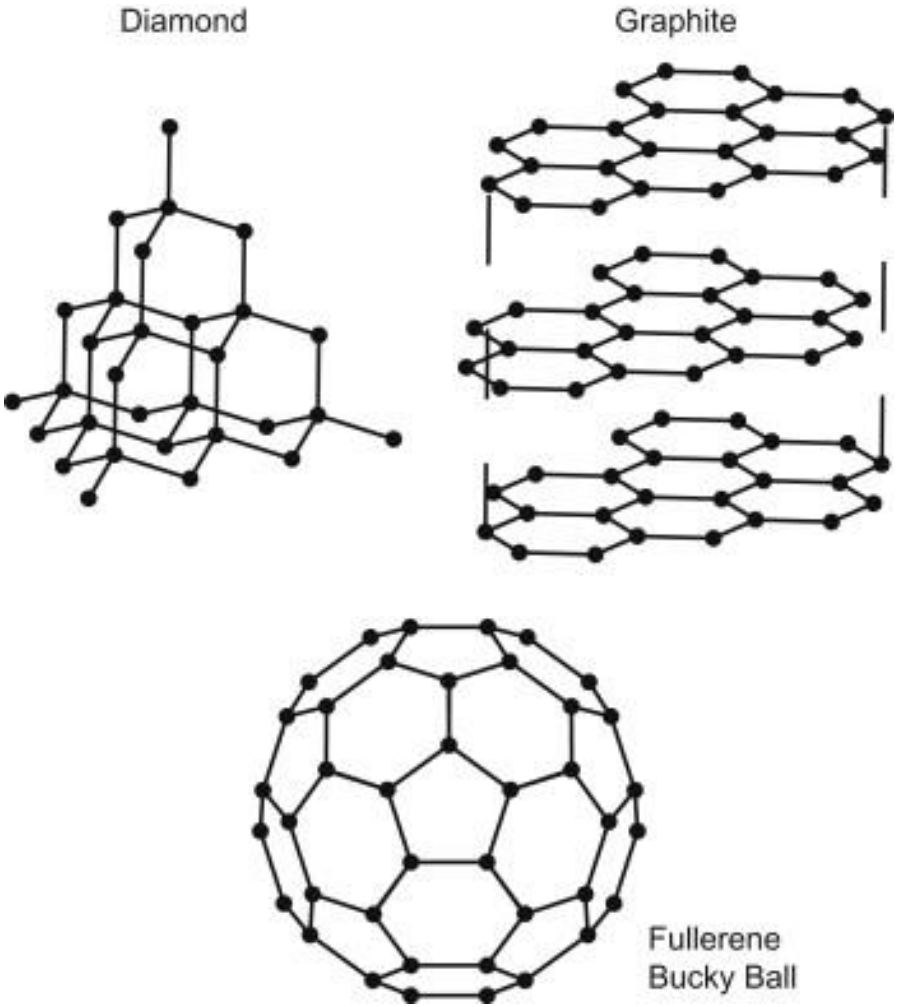
Chemistry:

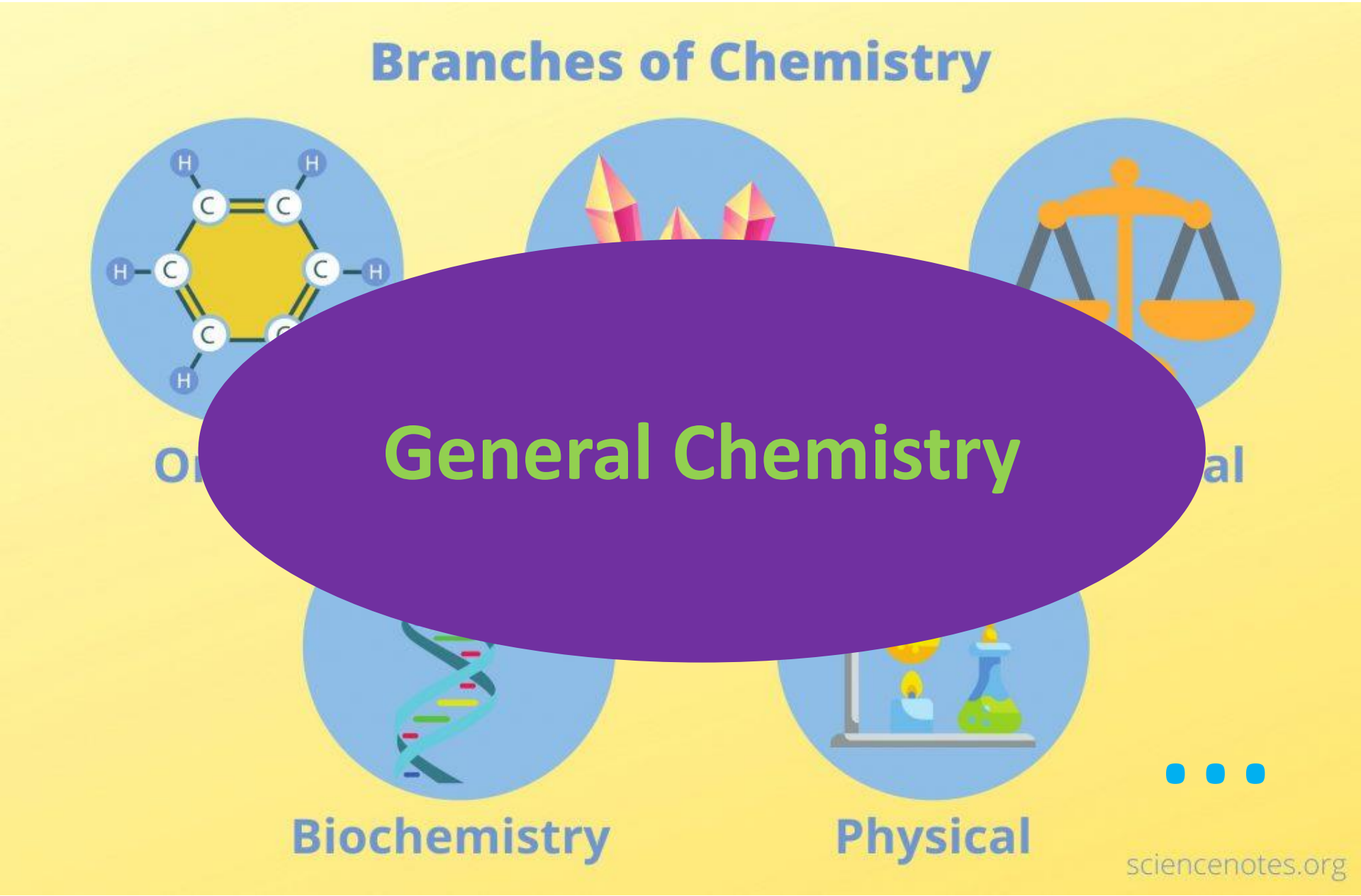
The study of composition, properties and reactivity of matter.

Matter

Chemical Changes

Physical Changes




Branches of Chemistry
General Chemistry
Biochemistry
Physical
sciencenotes.org

Summary questions

Over the total score of 20 points (General Chemistry I), the final exam will provide the maximal score of:

A) 4 points

B) 8 points

C) 10 points

D) 12 points

Summery questions

I don’t really like the General Chemistry I class. To be qualified for participating in the final exam, I should show up in at least:

A) 5 over 14 sections

B) 10 over 14 sections

C) 9 over 14 sections

D) 14 over 14 sections

Summery questions

The midterm exam of General Chemistry I will come:

A) After the 5th lecture section (Atoms: atomic structure, electronic structure)
B) After the 7th lecture section (Periodic properties of elements)
C) After the 10th lecture section (Bonding theories)
D) At the same time with the Final exam

Questions/ Comments?

